



TOMORROW IS TINY!

Let's set the scene: it's 2030 now, (or 2040, if you're reading this in 10 years time and none of these predictions have come true), and nanotechnology has really taken off. You, a time traveller from the primitive age of 2020 have come here to the future in order to escape the boring COVID-19 pandemic that forced you to sit at home and get so bored that you built an actual time machine! What is it you see around you that's different? Let's take it sector by sector, shall we?

Transport

The very first thing you notice is that all the cars on the roads are silent, and all gleaming like new! What's going on? Did everyone buy a new car today? After asking around you are told that the reason all the cars look so good is because they all have self-healing paint with the ability to buff out scratches automatically, as well as repair small dents. As for them being quiet, that's because in 2020 a pair of Indian youth from Bihar hit upon an idea to dedicate their lives to building a better battery, and using nanomaterials to store charge... they ended up increasing battery performance

and efficiency by 100 per cent! What this meant was that people were able to charge their cars in under half an hour, and each charge brought a range of 1,000 km, in Indian traffic! The multi-coloured spray on solar panels that were developed in Florida in 2023 meant that car batteries were being recharged in the daylight while they were being driven.

By 2026, carbon fibre manufacturing was so commonplace and cheap, that everything was made with it and polymers reinforced with it. The age of lightweight plastic cars that were stronger than metal was here to stay! Of course reduced weight means less downforce on the tyres, so to counter this, every single car has spoilers and the old supercar-style aerodynamic design. In fact, given the high taxes on combustion engines, and hard cap on engine capacity and emissions, only the tiniest of petrol cars will still exist in 2040, and will be so expensive to own and run, that only the truly rich will be able to afford all the penalties! We might actually live long enough to see the Tata Nano become one of the world's most expensive cars!

Inside engines too, lubricants filled with nano particles were helping reduce friction down to negligible amounts, which meant far better efficiency and torque. Nanocoatings meant that rusting was unheard of now, and windows everywhere (not just in cars) were tougher, totally shatterproof, and auto adjusted tint to let in just the perfect amount of light. They were also dust and water repellent, making wipers totally unnecessary – though of course laws take ages to catch up to technology, so wipers are still a legal requirement

on cars as of yet. Nanocoatings that make all surfaces totally free of static electricity means that dust and dirt just does not accumulate, either on our clothes or in our vehicles.

Public transport gets all of the same benefits as well, of course, but reduced friction and lightweight materials has reduced the cost of production of buses and trains, as well as the cost of running them. Cheaper and more efficient transport systems have resulted in much more reliability and comfort for travellers, which in turn also means more people use public transport. Even Mumbai's trains have got doors and airconditioning in all compartments, and have become standing room only in most carriages. The air filtration systems in the Mumbai locals is a feat of engineering, as teams from across the world worked on creating the most effective and efficient system known to mankind. Nanofilters are able to filter out odours and particulate matter to achieve 5 parts per trillion level of filtration, that too in a space jam-packed with people. What was interesting was the decision to suck air out of the compartment from the floor and lower walls, and pumping in filtered air from the top, which means that everyone is always breathing in fresh air.

Medicine

By far, the most advances by 2030 will have been made in medicine thanks to nanotechnology. While every generation thinks their medical technologies were breakthroughs and cutting edge, it will be the coming of nanotechnology that will truly give doctors the power to be gods and truly save lives as they want to.

Neural tissue interfaces

One of the most promising applications of nanotechnology is brain computer interfaces, and how nanotechnology promises to make more biocompatible electrodes. <http://dgit.in/nanolink>

Widespread nanotech

How would the world change if the nanotechnologies currently being developed is completely realised as products, how various sectors would transform, and what differences it would make to the end user. <http://dgit.in/nanoatom>

Future policy directions

India has a major role to play, and this paper explores the effects of globalisation on the nanoscale, with a particular focus on the future directions of policy in the US, China and India, because of the changes brought about by nanotechnology. <http://dgit.in/nanofuture>